

SynapseLab: Synaptic Plasticity as Short-Term Memory

DataForge 2026 • Pathway Track • One-Page Concept Summary

Central claim

AI systems can use changing synaptic connections as temporary memory, but conflicting information can interfere with that memory. SynapseLab makes this claim testable through a small interactive association task rather than a static explanation.

Problem and mechanism

Sequence models need a way to preserve information that is useful later. Transformers primarily use attention to route information among token representations, while their learned parameters normally remain fixed during inference. Synaptic-plasticity approaches introduce a different state variable: connection strengths can change temporarily as activity occurs. A simplified Hebbian update is $\mathbf{w}_{ij}(t+1) = \mathbf{w}_{ij}(t) + \eta \mathbf{x}_i(t) \mathbf{x}_j(t)$, where η controls the update size. Co-activation therefore leaves a temporary trace in connection state. This is an educational abstraction, not a complete biological model. [1,2]

What the learner changes

The SynapseLab experiment starts with an association such as **CAT** → **MEOW**. The learner can teach the same input a conflicting association, **CAT** → **BARK**, and observe the displayed memory strengths change. The visible state turns interference into something measurable: repeated competing evidence can reduce the relative dominance of an earlier association. The learner can therefore reproduce the claim, compare the output with the expected direction of change, and explain why the result occurred.

Connection to BDH

The concept is directly relevant to **Dragon Hatchling (BDH)**, a brain-inspired post-Transformer architecture. The published BDH work describes computation using locally interacting neuron particles and synapses and uses synaptic plasticity with Hebbian learning for working memory during inference. It reports that particular synapses can strengthen as particular concepts are processed. [3] SynapseLab translates this idea into a transparent educational substrate: **activity** → **synaptic update** → **changed future response**. Its BDH section is explicitly a toy/educational illustration, not an official BDH implementation or reproduction.

Architectural comparison

A conventional Transformer offers mature scaling, strong benchmark performance and efficient attention-based sequence processing, but its parameters are not normally modified for each new inference context. A plasticity-based recurrent system can store short-lived information in changing connection state; Kozachkov et al. (2022) found short-term synaptic plasticity could support robust, brain-like working memory in recurrent networks. [1] BDH goes further conceptually by combining neuron-synapse dynamics with attention-like computation in a post-Transformer design. [3] These are different design points, not evidence that BDH universally outperforms Transformers.

Evidence, trade-offs, and limitations

The evidence for synaptic plasticity as a computational memory mechanism includes controlled recurrent-network experiments and reviews of fast Hebbian plasticity. [1,2] BDH provides a published architectural example connecting synaptic plasticity to modern sequence-model design. [3] However, the evidence levels must be separated: a published benchmark is not a deployment, and SynapseLab's toy simulation is not a measurement of biological memory or the full BDH system. Temporary plasticity can introduce interference when competing associations share limited state. The simple model also omits many biological and architectural details such as decay, inhibition, normalization and richer plasticity rules. These limitations are part of the lesson rather than hidden from the learner.

What this artifact teaches

After the interaction, a learner should be able to distinguish transient activity from temporary synaptic state, describe a Hebbian-style update, reproduce interference with a conflicting association, and identify why synaptic plasticity matters in BDH. The artifact follows DataForge's required pattern of one precise claim, a meaningful manipulable variable, visible state, immediate feedback, a substantive BDH connection, and explicit evidence/limitation labels. [4]

Primary references

[1] Kozachkov, L. et al. (2022). *Robust and brain-like working memory through short-term synaptic plasticity*. PLOS Computational Biology, 18(12), e1010776. DOI: 10.1371/journal.pcbi.1010776.

[2] Lansner, A., Fiebig, F., & Herman, P. (2023). *Fast Hebbian plasticity and working memory*. Current Opinion in Neurobiology, 83, 102809. DOI: 10.1016/j.conb.2023.102809.

[3] Kosowski, A. et al. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*. arXiv:2509.26507.

[4] DataForge 2026 Pathway Track Problem Statement. Requirements for the interactive explainer, BDH/BDH-CQ module, evidence discipline, and concept summary.